# SAMS Kernel

Developing with a real TI: running, debugging, editing

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For a TI-99/4A with an F18A video card and a SAMS memory card
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Can you develop for the kernel *on* a TI? Partly, and this document says exactly how far, what is checked and what is not. The kernel's build is a PC program (Python and the xdt99 assembler), and the kernel itself loads from a cartridge into SAMS; both facts shape what a TI can do. Three levels (the third is new in 1.3.0):

| Level | What you do on the TI | Works today? |
|---|---|---|
| **A. The TI as the test machine** | run what the PC built, look at it, test it | yes (with the caveats below); the normal way |
| **B. The TI as the editor too** | write the source on the TI, move the files to the PC to build | yes; the sources are kept safe for it (part 4) |
| **C. Everything on the TI** | edit, assemble and run with no PC, in the workbench | **yes, for one-module programs, in the simulator** (part 5, and WORKBENCH.md); not on real hardware yet |

**Which route is which. Two ways to work: [PC] / [TI] / [PC or TI]. [PC]** is a PC with Python 3 and xdt99 (build with `skbuild.py`, run in the simulator or js99er); **[TI]** is the TI itself with the workbench cartridge (editor, assembler, launcher); **[PC or TI]** is the same either way. The full table is in PROGRAMMERS-GUIDE.md part 0. The three levels above: A and B are **[PC]** builds with the **[TI]** as the test machine or the editor; C is the **[TI]** route entirely.

**What is verified.** All of this project is verified in a simulator (a TMS9900, the F18A, SAMS, a keyboard, TI's own console ROM and disk controller ROM), not on a real TI-99/4A. No image from this project has yet been run on a real FinalGROM 99 or a real F18A and SAMS card. Treat the first run on real hardware as a test, and use the ladder in part 2.

## 1. What you need on the TI

A TI-99/4A with an **F18A** video card (or a PICO9918), a **SAMS** card of at least what the application asks for (512K by default), and a way to load a cartridge image: a **FinalGROM 99** or any cartridge that presents 8K banks selected by writes to `>6000 + 2n` (the images are in the "8" format: the build's output is named `NAME8.bin`). The F18A and SAMS are not optional: the loader refuses to start without them, and says so on the screen (part 2).

## 2. Level A: running on the TI, and the first-run ladder

Build on the PC (`python3 tools/skbuild.py APPDIR`), copy `build/APPDIR/NAME8.bin` to the cartridge's storage, and select it. For anything you will rely on, **climb the ladder**: each rung tells you something the one below cannot.

1. **The simulator and js99er.** The tests and a browser emulator (F18A and SAMS on) find nearly every programming mistake. They are not the machine: they do not model timing, and a real disk controller behaves differently from a stand-in in ways that matter (the file service found two: after the read that reports end of file TI's controller has already closed the file, and a missing file opens with error 2).
2. **`examples/hello`** (`images/hello8.bin`). Power on, select it, type: the screen is blue, a title bar, your typing echoed. If this works, the loader, SAMS, the F18A, the keyboard and the kernel all work.
3. **`examples/diag`** (`images/skdiag8.bin`). The key decoder (every chord shows its code and modifiers), the whole glyph set, the colour roles; **CTRL T** runs the kernel's self-test: hardware, paging, page pool, a module call, the keyboard, the screen routines, DMA, and a write, read and delete on DSK1 (it needs a disk in drive 1; a failing check shows its step and the DSR's error: step x 100 + error).
4. **`examples/modes`** (video), **`examples/music`** (sound: listen for the tune and the A), **`examples/browser`** (a catalog and a text file from your own disk).
5. **Your application.**

**What the loader says when it refuses.** It shows one of these in plain text and waits for FCTN = to return to the title:

| Message | What it means |
|---|---|
| NEEDS A SAMS MEMORY CARD WITH 512K OF MEMORY OR MORE TO RUN | no mapper answered at CRU `>1E00`, or the card has fewer pages than the application's `min_pages` (the size probe reads the pages it can tell apart) |
| THE MEMORY CARD FAILED ITS TEST AT PAGE >xx | a page from `>10` up read back what it was not given: the page number is the first that failed, so a 4K area of RAM (a chip or its socket) is bad |
| NEEDS AN F18A VIDEO CARD FOR ITS 80-COLUMN SCREEN AND COLOR | the F18A did not identify itself: not fitted, not connected, or its firmware does not answer the status register the loader reads |

If none of them appears and the screen stays dark, the loader got as far as starting the kernel; suspect the application (below).

## 3. Debugging on a TI that has no debugger

There is no monitor or single-stepper. What you have, and it is enough for most things:

- **`kernel/dbg.a99`: a routine that shows registers.** `COPY "dbg.a99"` in your module, then `BL @DBG` anywhere: the bottom row shows R0-R3 in hex and waits for a key; every register is kept, so you can put it between any two instructions. Load a register with what you want to see first (`MOV @SEL,R0`). It is tested in the simulator (`tests/test_kernel.py dbg`).
- **A heartbeat.** To see that the main loop is alive, show the frame counter in a corner from your `TICK`: `LI R0,COLS-5 / MOV @FRAME,R1 / LI R2,ABAR / BL @HEX4`. If it stops counting, your `TICK` (or something it calls) never returned: a loop with no exit, a stack overflow, or a call that lost its return address.
- **The self-test (CTRL T in `diag`)** after anything that touches paging, the stack or files.
- **Breadcrumbs.** Write a letter to a fixed cell as you pass milestones (`PUTC`); the last letter on a hung screen is where it stopped.
- **FCTN = quits** only if your `TICK` is running: a hung program needs the console's reset switch or power-off.

**Typical symptoms.** A blank screen after the refusal screens are gone: the module crashed in `INIT` (check you assembled to `AORG >C000` and kept the entry table first). A screen that stops updating but keys still beep: `TICK` runs too long (the frame counter). Garbled text colours in a mode with enhanced colour: you passed role attributes where flags belong (VIDEO-MODES.md part 5). A crash after a file call: you kept a value in `R6-R9` across it. Music stutters: your code runs long between `KIDLE` calls. VIDEO-MODES.md, PROGRAMMERS-GUIDE.md part 14 and CONVENTIONS.md have the rest.

## 4. Level B: editing on the TI

A TI text file is **DIS/VAR 80**: lines of 80 columns of ASCII. Any editor on the TI that writes that kind of file (TI-Writer, the Editor/Assembler's editor, a word processor that saves plain text) can edit the sources. Two things make it safe:

- **Every source line is at most 80 columns, plain ASCII, no tabs.** An editor cuts or splits a longer line; the shipped sources had 99 lines that would have been damaged (a string or a data list cut short is a silent bug). `tools/lint80.py` checks the rule, `tools/lint80.py --fix` repairs a file without changing a byte of the program (it moves a long comment above its line, re-wraps a long whole-line comment, splits a long string or data list; the build's images come out identical), and the test group `sources` checks every source in the project. Keep your own sources to it: run `lint80.py` before you copy them to the TI.
- **Moving files.** A disk image carries them both ways. xdt99's `xdm99` makes and reads TI disk images (tested here with `kernel/pool.a99`: out and back, the same text):

```
python3 xdm99.py -X sssd disk.dsk                        a blank single-sided single-density disk image
python3 xdm99.py disk.dsk -a main.a99 -n MAIN -f DIS/VAR80     put a source file on it, as the TI will see it: DSK1.MAIN
python3 xdm99.py disk.dsk                                its catalog
python3 xdm99.py disk.dsk -e MAIN -o main.a99            take a file back off
```

How the image gets to and from the real machine (a TIPI, a CF7+, a floppy written by a PC, a serial link) depends on your hardware; I have not tried any of them.

The loop is then: edit on the TI, move the files to the PC, `skbuild.py`, move the image to the cartridge, run. It works, and it is slow: do most of the iterating in the simulator and js99er, and use the TI to confirm.

## 5. Level C: everything on the TI, in the workbench

`examples/workbench` is an application on the kernel with an **editor**, a **native assembler** and a **launcher**: WORKBENCH.md is its manual. You write a program in the editor, press CTRL A, and run it with R, all on the TI. What is checked: the assembler's output is byte-identical to xas99's for every example module and two probe sources (every instruction and operand form), and the editor, the catalog and the run are exercised by key scripts, in the simulator.

What it is not, and what is still needed:

- **A PC once.** The workbench cartridge image and the disk holding `KERNEL/EQU` (the kernel's addresses) are made with `skbuild.py` and `skdisk.py`; they are prebuilt in `images/`. After that the PC is not needed to edit, assemble and run.
- **One-module programs, up to 4,096 bytes, run inside the workbench** (not a cartridge: a FinalGROM's storage cannot be written by the TI). To make a cartridge, build the program on the PC.
- **Not run on a real TI-99/4A**, and the time an assembly takes there is not measured (1.9 to 8.2 million instructions for the examples).
- **Not supported:** user macros (`.defm`), forward `EQU`, relocatable output, saving the assembled image to disk.

## 6. Differences between the TI and the simulator, to expect

| What | Where it is written down |
|---|---|
| a TI disk controller closes a file when a read reports the end, and answers 2 for a missing input file | PROGRAMMERS-GUIDE.md part 9 |
| the console's frame counter advances by the byte at `>83FC` | HARDWARE.md |
| timing: the simulator runs a fixed number of instructions a frame; a real TI does not | the simulator's `VSYNC_INSNS`; test anything timing-sensitive on the TI |
| video: the renderer follows js99er's model of the F18A, not the F18A | VIDEO-MODES.md |
| sound: the player's bytes and timing are verified; how it sounds is not | MUSIC.md |
| a real card's SAMS, a real F18A's firmware and a PICO9918's quirks | nothing verified |